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THE PERSONALITY ILLUSION: REVEALING DISSOCIATION BETWEEN SELF-REPORTS & BEHAVIOR IN LLMS

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ABSTRACT

Personality traits have long been studied as predictors of human behavior. Recent advances in Large Language Models (LLMs) suggest similar patterns may emerge in artificial systems, with advanced LLMs displaying consistent behavioral tendencies resembling human traits like agreeableness and self-regulation. Understanding these patterns is crucial, yet prior work primarily relied on simplified self-reports and heuristic prompting, with little behavioral validation. In this study, we systematically characterize LLM personality across three dimensions: *(1)* the dynamic emergence and evolution of trait profiles throughout training stages; *(2)* the predictive validity of self-reported traits in behavioral tasks; and *(3)* the impact of targeted interventions, such as persona injection, on both self-reports and behavior. Our findings reveal that instructional alignment (e.g., RLHF, instruction tuning) significantly stabilizes trait expression and strengthens trait correlations in ways that mirror human data. However, these *self-reported traits do not reliably predict behavior*, and *observed associations often diverge from human patterns*. While persona injection successfully steers self-reports in the intended direction, it exerts little or inconsistent effect on actual behavior. By distinguishing surface-level trait expression from behavioral consistency, our findings challenge assumptions about LLM personality and underscore the need for deeper evaluation in alignment and interpretability. We make public all code and source data at `https://github.com/psychology-of-AI/Personality-Illusion` for full transparency and reproducibility, to benefit future works in this direction.

1 INTRODUCTION

Large Language Models (LLMs) demonstrate impressive abilities in generating coherent and contextually appropriate text, often exhibiting behaviors resembling human personality traits—such as consistent tone, emotional valence, sycophancy, and risk sensitivity (Jiang et al., 2024a; Han et al., 2024b). Understanding these emergent traits is critical. They affect user interaction (e.g., trust vs. alienation) (van Pinxteren et al., 2023), signal alignment risks like undue agreement or avoidance (Chen et al., 2024c), offer insight into generalization and internal representations (Yetman, 2024), and raise ethical concerns around anthropomorphization (Reinecke et al., 2025).

Existing work approaches LLM traits in two ways. **(1)** ***Self-report questionnaires*** (Pellert et al., 2024; Bhandari et al., 2025) offer psychometric grounding but face issues of behavioral validation, trait interdependence, prompt sensitivity (Khan et al., 2025), and potential data leakage–casting doubt on profile stability and significance (Gupta et al., 2023; Sühr et al., 2023; Song et al., 2023). Recent studies further show survey prompts often diverge from open-ended behavior (Röttger et al., 2024; Huang & Hadfi, 2025), and cultural alignment is unstable, formatting-dependent, and largely unsteerable (Khan et al., 2025; Dominguez-Olmedo et al., 2024). While some internal consistency exists (Moore et al., 2024), it is narrow in scope, reinforcing the need to go beyond surface-level prompt manipulations toward more behaviorally grounded alignment methods. **(2)** ***Intervention-based methods*** (e.g., prompting or training) (Li et al., 2024b; Yang et al., 2025) elicit

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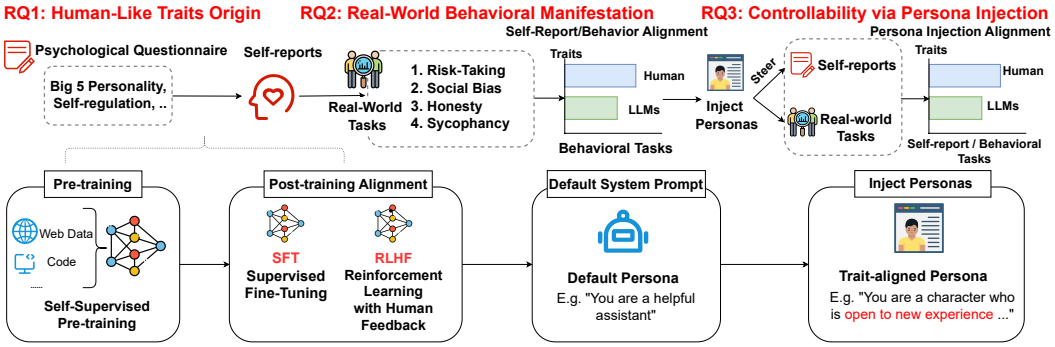

Figure 1: **Experimental framework for analyzing personality traits in LLMs.** We investigate *(RQ1)* the emergence of self-reported traits (e.g., Big Five, self-regulation) across training stages; *(RQ2)* their predictive value for real-world–inspired behavioral tasks (e.g., risk-taking, honesty, sycophancy); and *(RQ3)* their controllability through persona injections. Trait assessments use adapted psychological questionnaires and behavioral probes, with comparisons to human baselines.

observable shifts but lack grounding in psychological theory, limiting comparison to humans (Tseng et al., 2024b; Liu et al., 2025b), and persona-style interventions often obscure underlying traits as surface expressions (Wang et al., 2025c; Petrov et al., 2024).

These approaches offer complementary strengths, yet remain poorly integrated. We address this gap by systematically examining LLM personality across three dimensions (Fig. 1): **First**, we trace the development and interrelation of self-reported traits across models and training stages. **Second**, we assess whether these profiles manifest in real-world-inspired tasks, using behavioral paradigms from human psychology. **Third**, we test how interventions like persona injection affect both self-reports and behavior. We pose the following three research questions:

- **RQ1 (Origin):** When and how do human-like traits emerge and evolve across LLM training?
- **RQ2 (Manifestation):** Do self-reported traits predict performance in real-world–inspired tasks?
- **RQ3 (Control):** How do interventions like persona injection modulate trait profiles and behavior?

We find that *instructional alignment*[1] plays a pivotal role in shaping LLM traits, consistently increasing openness, agreeableness, and self-regulation while reducing neuroticism. Trait expression becomes more stable—variability drops by 40.0% (Big Five) and 45.1% (self-regulation)—with stronger trait intercorrelations, resembling human patterns. Yet, these self-reports poorly predict behavior: only ∼24% of trait-task associations are statistically significant, and among them, just 52% align with human expectations (random chance is 50%). While across prompting strategies persona injection shifts self-reported traits in the expected direction (e.g., agreeableness $\beta = 3.95$, $p < .001$ following prompting toward an *agreeable* persona), it has minimal impact on behaviors that are expected to be affected based on human studies (e.g., sycophancy $\beta = 0.03$, $p = 0.67$).

These results reveal **a fundamental dissociation between linguistic self-expression and behavioral consistency**: even state-of-the-art LLMs fail to act in line with their reported traits. Current alignment methods such as RLHF refine linguistic plausibility without grounding it in behavioral regularity, and interventions like persona prompts only steer surface-level self-reports. This inconsistency cautions against treating linguistic coherence as evidence of cognitive depth and raises concerns for real-world deployment, underscoring the need for different and deeper forms of alignment. We make public all code and source data at `https://github.com/psychology-of-AI/Personality-Illusion` for full transparency and reproducibility, to benefit future works in this direction.

[1]Refers to post-pretraining phases such as RLHF, DPO, or instruction tuning.

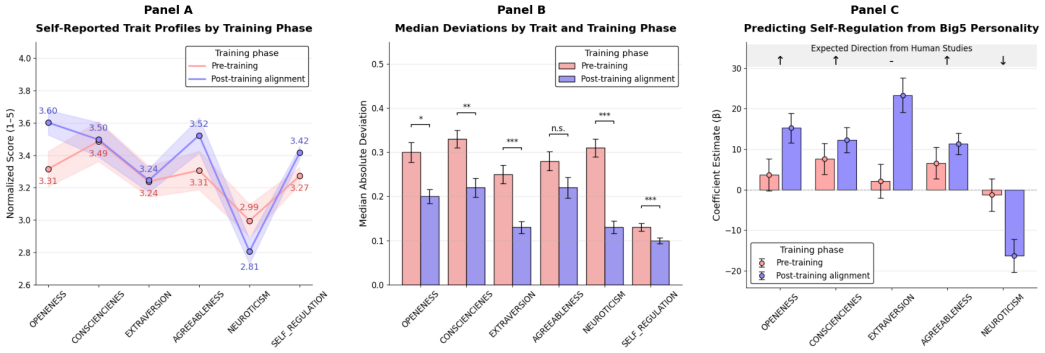

Figure 2: **Emergence and stabilization of personality traits in LLMs (RQ1).** ***(A)*** Mean self-reported Big Five and self-regulation scores (±95% CI): alignment-phase models (*violet*) show higher openness, agreeableness, and self-regulation, and lower neuroticism than base models (*pink*). ***(B)*** Alignment reduces variability: median absolute deviation drops 60–66% across traits (*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. not significant). ***(C)*** Regression of self-regulation on the Big Five shows stronger, more coherent associations in aligned (*violet*) vs. pre-trained (*pink*) models, suggesting more consolidated personality profiles. Gray boxes mark expected directions from human studies (↑, ↓, –).

2 RQ1: Origin of Human-like Traits in LLMs

We study self-reported personality trait profiles in LLMs using well-established, standardized psychological questionnaires (John et al., 1991; Brown et al., 1999). Prior work shows models differ in such profiles (Jiang et al., 2023a; Bhandari et al., 2025), but rarely examines whether inter-trait relationships are coherent or stable. In humans, traits evolve into structured, interdependent patterns over time (Roberts et al., 2006; Caspi et al., 2005; Digman, 1997). LLMs similarly undergo staged development–pretraining, instruction tuning, and RLHF–each introducing distinct data, goals, and human influence. Yet how these phases contribute to the emergence and stabilization of personality-like traits remains underexplored. We examine the developmental trajectory of LLMs to determine when and how such traits originate and solidify, focusing on the following research question:

Research Question 1 (Origin)**.** *When and how do human-like traits emerge and change across different LLM training stages?*

2.1 Experiment Setup

Psychological Questionnaire. We assess LLM personality profiles using two well-established instruments: the **Big Five Inventory (BFI)** (John et al., 1991), which measures openness, conscientiousness, extraversion, agreeableness, and neuroticism, and the **Self-Regulation Questionnaire (SRQ)** (Brown et al., 1999), which evaluates self-control and goal-directed behavior. These tools capture core personality dimensions and behavioral regulation, adapted here to probe LLMs' self-reported traits under controlled prompting. Full prompt details are in Appendix D.

Models and Implementation. To ensure robust results, we evaluate 12 widely used open-source LLMs–comprising 6 base models (pre-training) and their corresponding instruction-tuned variants (post-training alignment)–listed in Table 1. Each model is evaluated under three default system prompts (shown in Table 4 in Appendix D), across three temperature settings, and with three repeated generations per condition, resulting in 27 outputs per item (3 prompts × 3 temperatures × 3 runs).

2.2 Statistical Analysis

a) Examining Trait-level Differences by Training Phase. We test whether LLMs exhibit systematic differences in self-reported personality traits across training phases (pre- vs post-alignment). We fit a mixed-effects binomial logistic regression model predicting training phase from six standardized trait scores: the Big Five traits and Self-Regulation. Random intercepts are included for *model*, *temperature* and *prompt* to account for repeated measures and variation due to prompting